Page: Date: / /

FALL 2017-2018

Q1:

(a) $\frac{dy}{dx} = e^{-y}(1+x^2)$

$\int e^y dy = \int 1+x^2 dx \rightarrow e^y = x + \frac{1}{3}x^3 + c.$

(b) $z = \frac{y}{x} \rightarrow zx = y \rightarrow \frac{dz}{dx}x + z = \frac{dy}{dx}$

$x\frac{dz}{dx} + z = z^2 + 5z + 4$

$x\frac{dz}{dx} = z^2 + 4z + 4 \quad (z+2)^2$

$\int (z+2)^{-2} dz = \int \frac{1}{x} dx \Rightarrow -(z+2)^{-1} = \ln|x| + c.$

$-(\frac{y}{x}+2)^{-1} = \ln|x| + c.$

(c) $\frac{\partial M}{\partial y} = 2y \qquad \frac{\partial N}{\partial x} = 2y \quad \therefore$ Exact

$\int M dx + \int N dy = C$

$xy^2 + \frac{1}{3}x^3 + xy^2 = C \Rightarrow \boxed{xy^2 + \frac{1}{3}x^3 = C}$ #.

(d) $\mu = e^{\frac{1}{2}\int \frac{1}{x}(-4)} = e^{2\ln x} = e^{\ln x^2} = x^2$

$y^{-4} = x^{-2}\left[\int x^2 \cdot 5x^2 \cdot (-4)\, dx + c\right] = x^{-2}\left[\frac{-20}{5}x^5 + c\right]$

$= -4x^3 + \frac{c}{x^2}.$

Q2: $y''' - 4y'' + 5y' - 2y = 0$

$m^3 - 4m^2 + 5m - 2 = 0$

$m_1 = 2 \quad m_2 = 1 \quad m$

$y = C_1 e^{2x} + C_2 e^{x}$

الممسوحة ضوئيا بـ CamScanner

Page: Date: / /

(b) $y'' + 3y' - 10y = 2e^{3x}$

$m^2 + 3m - 10 = 0$

$(m-2)(m+5)$

$m_1 = 2 \quad m_2 = -5$

$y_{C.F} = C_1 e^{2x} + C_2 e^{-5x}$

$y_{P.I} = Ae^{3x}$

$y' = 3Ae^{3x}$

$y'' = 9Ae^{3x}$

$9Ae^{3x} + 9Ae^{3x} - 10Ae^{3x} = 2e^{3x}$

$8A = 2 \Rightarrow A = \frac{2}{8} = \frac{1}{4}$

$y_{P.I} = \frac{1}{4} e^{3x}$

$y_{G.S} = C_1 e^{2x} + C_2 e^{-5x} + \frac{1}{4} e^{3x}$

(c) $y'' + y = 0$

$m^2 + 1 = 0$

$m_{1,2} = \pm i$

$y = e^{0x}[C_1 \cos x + C_2 \sin x]$

$u = \cos x \qquad v = \sin x$

$u' = -\sin x \qquad v' = \cos x$

$uv' - vu' = 1$

$Ax(u) = \int \frac{-v f(x)}{1} = \int -\sin x \cdot \frac{\sin x}{\cos x} = -\int \frac{\sin^2 x}{\cos x}$

$= -\int \frac{1 - \cos^2 x}{\cos x} = -\int \frac{1}{\cos x} - \cos x$

$= -[\sec x \tan x - \sin x]$

$= -\sec x \tan x + \sin x + C.$

$Bx(v) = \int \frac{u f(x)}{1} = \int \cos x \cdot \frac{\sin x}{\cos x}$

$= \int \sin x = -\cos x + k.$

$y_{G.S} = e^{0x}[C_1 \cos x + C_2 \sin x] - \sec x \tan x + \sin x + C - \cos x + k$

(d) $y'' - 3y' + 2y = e^x$

$y = ve^{2x}$

$y' = \frac{dv}{dx} e^{2x} + 2e^{2x} v$

$y'' = \frac{d^2v}{dx^2} e^{2x} + 2e^{2x} \frac{dv}{dx} + 4e^{2x} v + \frac{dv}{dx} 2e^{2x}$

$\frac{d^2v}{dx^2} e^{2x} + 4e^{2x} \frac{dv}{dx} + 4e^{2x} v - 3e^{2x} \frac{dv}{dx} - 6e^{2x} v + 2e^{2x} = e^x$

$\frac{d^2v}{dx^2} e^{2x} + e^{2x} \frac{dv}{dx} = e^x \quad (\div e^{2x})$

$\frac{dv}{dx} = z \Rightarrow \frac{d^2v}{dx^2} = \frac{dz}{dx}$

$\frac{d^2v}{dx^2} + \frac{dv}{dx} = e^{-x}$

$\frac{dz}{dx} + z = e^{-x}$

CamScanner الممسوحة ضوئيا بـ

Page: Date: / /

$\mu = e^{\int 1\,dx} = e^{x}$

$y = e^{-x}\left[\int e^{x}\cdot e^{-x}\,dx + c\right] = e^{-x}\left[x + c\right] = xe^{-x} + e^{-x}c$

$\frac{dv}{dx} = xe^{-x} + e^{-x}c \rightarrow \int dv = \int xe^{-x} + e^{-x}c\,dx$

x	e^{-x}
1	$-e^{-x}$
0	e^{-x}

$v = -xe^{-x} - e^{-x} + e^{-x}c + k$

$\frac{y}{e^{x}} =$

$\# \; y = -xe^{x} - e^{x} + e^{x}c + e^{2x}k \; \#$

(23)

(a) $s^2 y(s) - s y(0) - y'(0) - 6s y(s) - 6 y(0) + 8 y(s) = \frac{1}{s-3}$

$y(s)\left[s^2 - 6s + 8\right] = \frac{1}{s-3} + \frac{2}{1} = \frac{1 + 2s - 6}{s-3}$

$y(s) = \frac{2s-5}{(s-3)(s-4)(s-2)} = \frac{A}{s-3} + \frac{B}{s-4} + \frac{C}{s-2}$

$A = -1$ $\quad B = \frac{3}{2}$ $\quad C = \frac{-1}{2}$

$y(s) = -\frac{1}{s-3} + \frac{3}{2}\frac{1}{s-4} + \left(\frac{-1}{2}\right)\frac{1}{s-2}$

$y(t) = -e^{3t} + \frac{3}{2}e^{4t} - \frac{1}{2}e^{2t}$

الممسوحة ضوئيا بـ CamScanner

Page: Date: / /

(b) (i) $f(s) = \frac{5!}{s^6} + \frac{7}{s-6} + \frac{8s}{s^2-4}$

(ii) $f(s) = \frac{1}{s}\left[\frac{4}{(s-1)^2+16}\right]$

(iii) $f(s) = e^{-3s}\left[\frac{1}{2s} - \frac{1}{2}\frac{s}{s^2+4}\right]$

(iv) $\int_s^\infty \frac{1}{z^2+1}\,dz \Rightarrow \tan^{-1} z \Big|_s^\infty = \infty - \tan^{-1} s$

Q4:

(a) (i) $F(s) = \ln(s^2+16) - 2\ln(s^2-1)$

$F'(s) = \frac{2s}{s^2+16} - \frac{2}{(s-1)}$

$-t f(t) = 2\cos 4t - 2e^t$

$f(t) = \frac{2\cos 4t - 2e^t}{-t}$

(ii) $f(s) = \frac{s}{s^2+10s+8+25-25} = \frac{s}{(s+5)^2-17} = \frac{1}{\sqrt{17}} e^{-5t} \cosh \sqrt{17}t$

(iii) $f(t) = 3u_4(t)\cos\sqrt{11}(t-4)$

الممسوحة ضوئيا بـ CamScanner